

- ENN removes misclassified instances near the decision boundary from the majority class.
- The sample code is:

```
from collections import Counter
from sklearn.datasets import make_classification
from sklearn.utils import resample
from imblearn.under_sampling import AllKNN
from matplotlib import pyplot
from numpy import where

X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)

# summarise class distribution
print('Original dataset shape %s' % Counter(y))
Original dataset shape Counter({1: 894, 0: 106})

# perform downsampling
allknn = AllKNN()
X_Res, y_res = allknn.fit_resample(X, y)
print('Resampled dataset shape %s' % Counter(y_res))
Resampled dataset shape Counter({1: 872, 0: 106})

# scatter plot of examples by class label
for label, _ in Counter(y_res).items():
    row_ix = where(y_res == label)[0]
    pyplot.scatter(X_Res[row_ix, 0], X_Res[row_ix, 1],
label=str(label))
pyplot.legend()
pyplot.show()
```

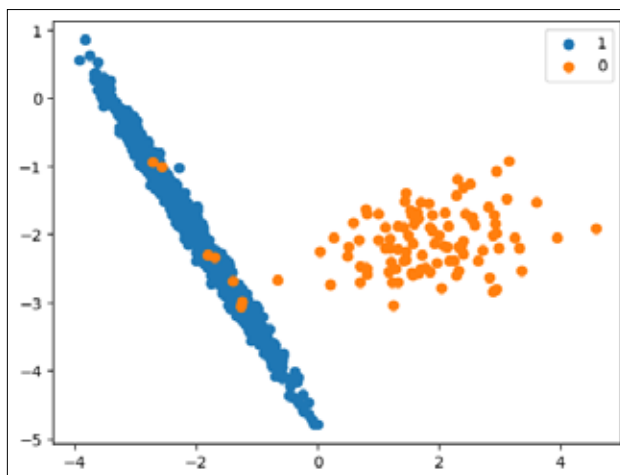


Figure 8: Scatter plot of examples by class label downsampling post using AllKNN

Pros and cons of downsampling

Pros

- Helps in reducing the bias where the class is over-populated.
- Reduces noise and improves accuracy.
- Saves training time and reduces the computational power needed.

Cons

- Can result in information loss.
- Post downsampling, the remaining dataset can lead to overfitting or biased results.

To sum up, upsampling involves an increase in the minority class while downsampling is vice-versa – it involves a decrease in the majority class. These equalisation processes prevent the machine learning model from inclining towards the majority class. Remember that upsampling and downsampling need to preserve the distribution of the data, and the boundary line between the target classes must remain the same. Also, upsampling and downsampling are applied only to the training datasets and no changes are made to validation and testing data. **END** 🐧

References

- https://imbalanced-learn.org/stable/references/over_sampling.html
- https://imbalanced-learn.org/stable/references/under_sampling.html
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The author works in a leading bank as AVP. She is a perennial learner and loves the quote: "The more I learn, the more I realise how much I don't know." - Albert Einstein